

JPHG01 Temperature and thermometer — Quiz 1

S3 Physics · 3 multiple-choice items

Q1 · JPHG01.3 · Foundation

Which of the following thermometers measures temperature by detecting the intensity of radiation?

- A. Liquid-in-glass thermometer
- B. Resistance thermometer
- C. Thermistor thermometer
- D. Infra-red thermometer

Q2 · JPHG01.1 · Foundation

Which of the following statements about the Celsius temperature scale is/are correct?

- (1) The lower fixed point is the lowest temperature existing in nature.
- (2) The upper fixed point is the temperature of steam.
- (3) The unit of this scale is $^{\circ}\text{C}$.

- A. (1) only
- B. (3) only
- C. (1) and (2) only
- D. (2) and (3) only

Q3 · JPHG01.2 · Standard

A faulty thermometer with uniform scale reads 5°C and 95°C when it is placed in melting ice and boiling water respectively. What is the actual temperature if the thermometer reads 30°C ?

- A. 27.8°C
- B. 28.5°C
- C. 30°C
- D. 35°C

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Q1 · JPHG01.3

Answer: D. Infra-red thermometer

Guide: The intensity of infra-red radiation given out by an object increases when its temperature increases.

Q2 · JPHG01.1

Answer: D. (2) and (3) only

Guide: The lower fixed point is the ice point, not the lowest temperature in nature. The upper fixed point is the steam point. The unit is °C.

Q3 · JPHG01.2

Answer: A. 27.8 °C

Guide: Ice point is 0 °C and steam point is 100 °C. $T/100 = (30 - 5)/(95 - 5)$.